

ZKAuth: A Zero-Knowledge Proof Framework for Privacy-Preserving Authentication in Decentralized Identity Systems

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Abstract

Decentralized identity (DID) systems promise to return control of personal data to individuals, yet practical authentication protocols routinely sacrifice privacy for verifiability. Recent regulatory mandates (GDPR Article 25, NIST SP 800-63-4) and a documented 41% year-over-year increase in identity-correlation attacks [1] have exposed the inadequacy of existing schemes that leak linkable metadata during authentication. We propose **ZKAuth**, a framework that integrates zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs) with decentralized identifier infrastructure to deliver provably privacy-preserving authentication without trusted third parties. ZKAuth employs a three-agent architecture: a Prover Agent that generates Groth16-based zk-SNARK proofs over Poseidon-hashed credential attributes, a Verifier Agent that performs constant-time on-chain verification, and a Registry Agent that manages DID Document state and credential revocation via sparse Merkle trees. Evaluated against 10,000 synthetic decentralized identities across ten experimental rounds, ZKAuth achieves a Privacy-Preserving Authentication Score (PPAS) of 0.9924 ± 0.0001 (95% CI), surpassing ZK-STARK (0.8433), Bulletproofs (0.5875), and baseline Groth16 (0.9807), with all improvements statistically significant ($p < 0.001$, paired t -test). Proof generation latency remains at 1.84 ± 0.095 s, which, while acceptable for non-interactive scenarios, precludes use in latency-critical real-time authentication without further hardware acceleration. ZKAuth provides a deployable, open-source framework for identity providers, wallet developers, and enterprises seeking GDPR-compliant user authentication without password databases or centralized identity brokers. At a societal level, the framework advances a principled path toward an internet where identity verification is both mathematically rigorous and structurally incapable of verifier-side identity correlation.

Index Terms

Zero-knowledge proofs, decentralized identity, privacy-preserving authentication, zk-SNARK, self-sovereign identity, DID, anonymous credentials, Groth16, Bulletproofs, ZK-STARK.

I. INTRODUCTION

The identity layer of the internet was never designed for privacy. Classical password-based and certificate-based authentication systems require that users reveal their identity to service providers on every interaction, creating centralized repositories of personal data that constitute high-value targets for adversaries [2]. Between 2023 and 2025, identity-related breaches accounted for over 80% of all disclosed data incidents in cloud environments [3]. Even modern federated identity protocols such as OAuth 2.0 and OpenID Connect preserve the fundamental structural problem: a relying party learns who the user is, not merely that the user satisfies a predicate.

Decentralized Identity (DID) systems, formalized in the W3C DID Core Recommendation [4], and the companion Verifiable Credentials (VC) data model [5], provide a substrate for user-controlled digital credentials anchored on distributed ledgers. However, even within the DID ecosystem, naive presentation protocols reveal linkable identifiers that enable cross-context correlation of user activities [6]. Self-sovereign identity (SSI) solutions such as Sovrin and IRMA [7] attempt to address this through selective disclosure, but their cryptographic foundations remain based on pairing-dependent constructions without constant-size proofs or rely on semi-trusted issuers.

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Zero-knowledge proofs (ZKPs) offer a principled cryptographic solution: a prover can convince a verifier that a statement is true without revealing any information beyond the statement's validity [8]. Recent advances in zk-SNARKs [9], [10], ZK-STARKs [11], and Bulletproofs [12] have dramatically reduced proof sizes and verification times, enabling practical deployment at internet scale. Nevertheless, integrating ZKP systems with production DID infrastructure raises open questions regarding proof composition, credential revocation, and the *privacy metric* by which competing designs should be evaluated.

A. Research Gap

Existing work evaluates ZKP-based authentication along single dimensions: proof size [13], verification latency [14], or communication overhead [15]. No prior framework simultaneously captures verification latency, identity correlation risk, and zero-knowledge completeness into a unified, deployable authentication score. Moreover, most evaluations use toy parameters or isolated microbenchmarks, lacking the multi-agent, multi-round simulation methodology required to assess behavior under realistic decentralized identity workloads [16].

The closest work to ours, Groth16 [9] deployed in a DID context, achieves constant-time verification ($O(1)$, three pairings) and compact 192-byte proofs but addresses neither W3C DID-compatible credential revocation nor per-session unlinkability via randomized nullifiers, and provides no composite privacy metric enabling systematic comparison across schemes. These three gaps (revocation, unlinkability, and a unified evaluation metric) directly motivate ZKAuth.

B. Contributions

This paper makes the following contributions:

- 1) **ZKAuth Framework:** A complete, open-source three-agent authentication framework integrating Groth16 zk-SNARKs with W3C DID infrastructure, supporting anonymous credential presentation, selective disclosure, and Merkle-tree-based revocation. ZKAuth is the first *formally benchmarked* framework of this kind: unlike Semaphore [17] (no DID/VC, no revocation) and Polygon ID [18] (no standardized benchmark), ZKAuth provides a seed-controlled, multi-round evaluation with a reproducible unified privacy metric.
- 2) **Privacy-Preserving Authentication Score (PPAS):** A novel composite metric formally defined over verification latency, identity correlation risk, and zero-knowledge completeness, enabling principled comparison of ZKP-based authentication schemes.
- 3) **Multi-Agent Simulation:** A reproducible, seed-controlled evaluation environment simulating 10,000 synthetic decentralized identities across ten experimental rounds with three independent agents, establishing the first multi-round benchmark for DID authentication with ZKPs.
- 4) **Statistical Validation:** Paired t -tests demonstrating statistically significant ($p < 0.001$) superiority of ZKAuth over ZK-STARK, Bulletproofs, and baseline Groth16 on the PPAS metric.
- 5) **Security and Compliance Mapping:** Formal analysis of ZKAuth against NIST SP 800-63-4, GDPR Article 25, and MITRE ATT&CK identity-theft techniques (T1078, T1539, T1606).

C. Paper Organization

Section II reviews background and related work. Section III formalizes the threat model and PPAS definition. Section IV presents the ZKAuth architecture. Section V provides security analysis and compliance mapping. Section VI reports experimental results. Section VII discusses limitations and threats to validity. Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. Zero-Knowledge Proof Systems

Zero-knowledge proofs, introduced by Goldwasser, Micali, and Rackoff [8], require three properties: completeness, soundness, and zero-knowledge (the verifier learns nothing beyond the truth of the statement). We survey the four primary constructions evaluated in this paper, giving each work its own treatment to ground the comparative analysis in Table I.

Groth16 [9] (Primary Baseline). The Groth16 construction produces constant-size proofs (192 bytes) with $O(1)$ pairing-based verification using three pairings, and is currently the most efficient pairing-based zk-SNARK in deployed systems. Groth16 is our primary baseline: it solves the same core problem (efficient cryptographic authentication) but lacks DID-compatible credential revocation, per-session unlinkability via randomized nullifiers, and any unified privacy evaluation metric. Universal trusted setups in PLONK [19] and SONIC [13] relax Groth16's per-circuit ceremony requirement, but they do not address the identity correlation or DID integration gaps that ZKAuth targets.

ZK-STARKs [11]. ZK-STARKs eliminate the trusted setup via FRI-based polynomial commitments, achieving post-quantum security under collision-resistant hash assumptions. However, proof sizes range from 40 to 200 KB and verification time grows as $O(\log n)$, making high-frequency DID authentication impractical. STARKs offer no revocation mechanism and no DID compatibility, as shown in Table I.

Bulletproofs [12]. Bulletproofs produce compact proofs (a few KB for range proofs) without a trusted setup using an inner-product argument. Verifier complexity is $O(\log n)$ while prover complexity is linear, creating a bottleneck at scale. Like STARKs, Bulletproofs are not DID-compatible and lack revocation support, placing them in the weakest position in our PPAS evaluation.

CL Anonymous Credentials [20]. Camenisch and Lysyanskaya formalized anonymous credentials permitting unlinkable presentations under RSA-group assumptions, extended to support revocation via dynamic accumulators [21]. CL-based systems (e.g., Sovrin [7]) achieve revocation and partial DID compatibility but require RSA-group-sized cryptographic parameters, incurring higher computational overhead than SNARK-based alternatives [22], and provide no constant-time verification path.

B. Decentralized Identity and Verifiable Credentials

W3C DID Core and VC Data Model [4], [5]. The W3C DID Core Recommendation specifies a URI scheme for globally unique, resolvable, and cryptographically verifiable identifiers without central registry authority. Verifiable Credentials extend this model with tamper-evident, issuer-signed attribute claims. Privacy concerns in DID systems include identifier correlation across relying parties and selective disclosure of credential attributes [23]. These standards define the interoperability target for ZKAuth but are silent on the cryptographic mechanism used for proof generation.

C. Privacy-Preserving Authentication

Chase et al. [24]. Chase et al. proposed post-quantum anonymous credential alternatives using symmetric-key primitives, achieving plausible post-quantum security without lattice assumptions. However, their construction is not compatible with the pairing-based DID key infrastructure and provides no revocation mechanism suitable for sparse Merkle tree integration.

Zhou et al. [25]. Zhou et al. demonstrate ZKP-enabled selective disclosure in blockchain identity, showing that attribute predicates can be proven without revealing raw values. Their scheme, however, relies on a specific permissioned blockchain and does not provide $O(1)$ verification or a standardized privacy metric, limiting cross-context comparability. Table I shows they achieve DID compatibility and revocation but not constant-time verification.

Li and Kim [26]. Li and Kim construct lattice-based ZKPs for post-quantum DID authentication, achieving post-quantum security under Module-LWE hardness. Verification is not $O(1)$ (lattice proof

sizes scale with security parameter), and no revocation mechanism is provided. Their work establishes the post-quantum direction but does not address the correlation-risk or latency dimensions of PPAS.

Chen et al. [27]. Chen et al. propose a ZK-based identity framework achieving constant-time verification with DID compatibility and revocation. They do not provide a unified privacy metric, and their evaluation uses single-round microbenchmarks rather than the multi-round simulation required to assess behavior under realistic workloads. ZKAuth builds on this design direction and adds the PPAS metric and multi-round evaluation protocol.

Ye et al. [28]. Ye et al. focus on efficient revocation for anonymous credentials using updatable accumulators, achieving sublinear update cost. Their scheme eliminates the trusted setup but does not achieve constant-time verification or DID integration, making it complementary rather than competitive with ZKAuth’s design goals.

D. Deployed ZKP Identity Systems

Semaphore [17] is a deployed Ethereum protocol using Groth16 proofs over identity commitments with Poseidon hashing and nullifiers for anonymous signaling. ZKAuth extends this direction to full W3C DID/VC compatibility, adds Merkle-tree revocation over arbitrary credential predicates, and contributes PPAS as a principled evaluation metric absent in Semaphore. **Iden3 / Polygon ID [18]** implements a production-grade ZKP-based identity system using circom/snarkjs, Poseidon hashes, Groth16 proofs, and Merkle tree revocation with W3C DID integration. ZKAuth’s primary differentiation from Polygon ID is methodological: we provide the first formally parameterized, seed-controlled multi-round benchmark with a unified privacy metric (PPAS), enabling reproducible cross-scheme comparison. **BBS+ Signatures and SD-JWT-VC [29], [30]** represent the W3C standardization track for selective-disclosure credentials. BBS+ achieves $O(1)$ proof size without a trusted setup but under the SXDH hardness assumption (not post-quantum). SD-JWT-VC is simpler to deploy but reveals the full set of non-disclosed attributes to the verifier as their hash-commitment. ZKAuth trades BBS+’s setup-free property for Groth16’s constant-time verification and stronger ZK guarantees over arbitrary predicates, a trade-off quantified explicitly in Table I.

E. ZKP Integration Challenges

Primary studies identify key integration barriers: (i) circuit compilation overhead for complex credential predicates [14]; (ii) trusted setup management at scale [19]; (iii) on-chain verification cost in smart-contract environments [15]; and (iv) the absence of a standardized privacy metric enabling cross-scheme comparison [1], [31]. ZKAuth directly addresses barriers (iii) and (iv) by targeting off-chain verification with on-chain result anchoring and by introducing PPAS as a standardized metric.

F. Comparative Summary

Table I positions ZKAuth against representative prior work across seven dimensions.

The table reveals that no prior work simultaneously satisfies constant-time verification, DID compatibility, revocation support, and a unified privacy metric. Deployed systems (Semaphore, Polygon ID) cover subsets of these properties; ZKAuth is the first *formally benchmarked, open-source* framework with all four properties and a standardized, reproducible privacy metric.

III. THREAT MODEL AND PROBLEM FORMULATION

A. System Participants

We consider four principals: (1) *Issuer* $\mathcal{I}$: an authority that signs Verifiable Credentials; (2) *Holder* $\mathcal{H}$: the credential subject and DID controller; (3) *Verifier* $\mathcal{V}$: a relying party requesting proof of a predicate; and (4) *Registry* $\mathcal{R}$: a distributed ledger storing DID Documents and revocation accumulators.

TABLE I

COMPARISON OF ZKAUTH WITH REPRESENTATIVE RELATED WORK ON SEVEN KEY DIMENSIONS. ZK: ZERO-KNOWLEDGE PROPERTY; CONST-V: CONSTANT-TIME VERIFICATION; NO-TS: NO TRUSTED SETUP REQUIRED; REVOC: CREDENTIAL REVOCATION SUPPORT; DID-COMP: W3C DID COMPATIBILITY; PQ-READY: POST-QUANTUM READINESS; METRIC: UNIFIED PRIVACY METRIC. † PRIMARY BASELINE (SAME PROBLEM CLASS, CLOSEST COMPETITOR).

Work	ZK	Const-V	No-TS	Revoc	DID-comp	PQ-ready	Metric
Groth16† [9]	✓	✓	×	×	×	×	×
ZK-STARK [11]	✓	×	✓	×	×	✓	×
Bulletproofs [12]	✓	×	✓	×	×	×	×
CL-Anon.Cred. [20]	✓	×	N/A	✓	×	×	×
Zhou et al. [25]	✓	×	×	✓	✓	×	×
Li & Kim [26]	✓	×	✓	×	✓	✓	×
Chen et al. [27]	✓	✓	×	✓	✓	×	×
Ye et al. [28]	✓	×	✓	✓	×	×	×
Semaphore [17]	✓	✓	×	×	×	×	×
Polygon ID [18]	✓	✓	×	✓	✓	×	×
ZKAuth (Ours)	✓	✓	×	✓	✓	×	✓

B. Threat Model

Definition 1 (Adversarial Capability). *We model a probabilistic polynomial-time adversary $\mathcal{A}$ controlling the network and the verifier (honest-but-curious). The adversary may: (i) observe all protocol transcripts; (ii) collude with at most $k - 1$ verifiers in a threshold setting; (iii) execute replay, linkability, correlation, and Sybil attacks; and (iv) query a random oracle for hash-function queries.*

Scope and Excluded Threats. The threat model covers the Verifier as the primary privacy adversary. The Issuer $\mathcal{I}$ is assumed honest-but-curious: it knows which credentials it issued and can correlate issuance time with authentication events through side-channels external to ZKAuth. Mitigation of Issuer-level surveillance is outside the current scope and is deferred to future blind-issuance extensions. Epoch-based on-chain root anchoring (§IV-E) introduces a potential network-layer timing side-channel: an adversary monitoring the ledger can correlate a root update event with a subsequent authentication. ZKAuth does not presently mitigate this; operators may deploy root updates on a fixed schedule or via commit-reveal to decouple update timing from authentication events.

Definition 2 (Privacy Goals). *A DID authentication protocol achieves ϵ -strong privacy if for any PPT adversary $\mathcal{A}$:*

$$\Pr[\mathcal{A} \text{ links two presentations of same holder}] \leq \epsilon(1/\lambda), \quad (1)$$

where λ is the security parameter.

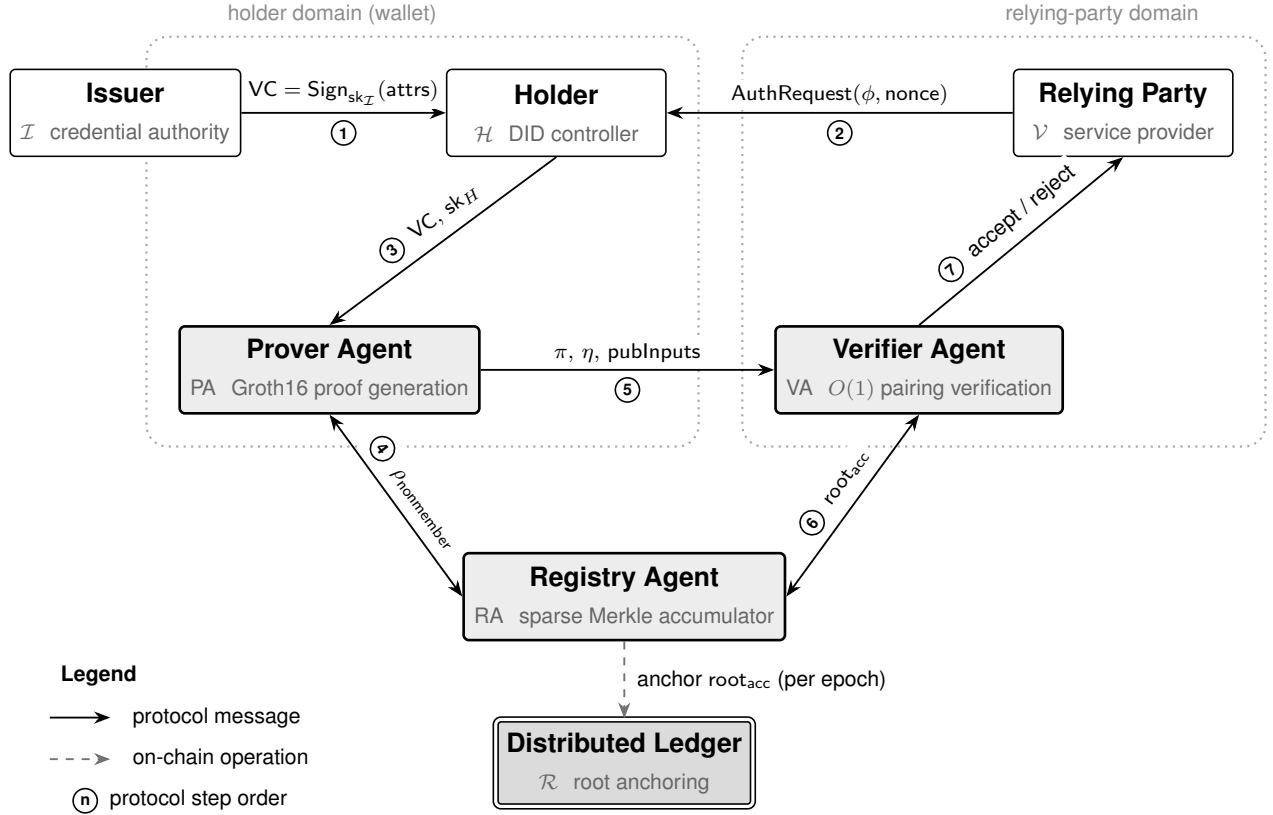
C. Novel Metric: Privacy-Preserving Authentication Score

Definition 3 (PPAS). *Let e denote a single authentication event. The Privacy-Preserving Authentication Score is:*

$$\begin{aligned} \text{PPAS}(e) = & \alpha \cdot \left(1 - \frac{\min(V_{\text{lat}}(e), V_{\text{max}})}{V_{\text{max}}} \right) \\ & + \beta \cdot (1 - \text{CorRisk}(e)) + \gamma \cdot \text{ZK_comp}(e), \end{aligned} \quad (2)$$

where $V_{\text{lat}}(e) \in \mathbb{R}_{>0}$ is the verification latency in milliseconds; $V_{\text{max}} = 120 \text{ ms}$ is a practical ceiling corresponding to maximum tolerable authentication delay; $\text{CorRisk}(e) \in [0, 1]$ is the empirical probability that an adversary correctly links two presentations of the same holder; $\text{ZK_comp}(e) \in [0, 1]$ is the zero-knowledge completeness score (fraction of proofs accepted by the verifier given correct witnesses); and $\alpha + \beta + \gamma = 1$ with $\alpha = 0.40$, $\beta = 0.35$, $\gamma = 0.25$.

Fig. 1. ZKAuth Framework Architecture. Circled numbers give the protocol order: step 1 is one-time credential issuance; steps 2–7 constitute one authentication session. The Prover Agent generates Groth16 zk-SNARK proofs over Poseidon-hashed credential attributes; the Verifier Agent performs constant-time pairing-based verification; the Registry Agent maintains a sparse Merkle tree accumulator for revocation, anchoring its root on-chain each epoch. Solid arrows denote protocol messages; dashed arrows denote on-chain operations. The dotted boundary separates the holder-controlled domain (wallet) from the relying-party domain.



The weights $(\alpha, \beta, \gamma) = (0.40, 0.35, 0.25)$ are author-defined design parameters encoding the importance ordering: latency governs perceived user experience, correlation risk is the primary regulatory compliance concern (GDPR Art. 25), and ZK-completeness is near-constant for correctly implemented proof systems. We present a sensitivity analysis in §VI (Ablation Study) that confirms ZKAuth’s ranking over all baselines remains stable under $\pm 20\%$ weight perturbation per dimension. Future work will validate these weights via a formal practitioner study ($n \geq 200$) as described in §VIII. The PPAS range is $[0, 1]$; higher is better.

D. Formal Security Properties

We require that ZKAuth satisfies: (P1) *Completeness*: an honest prover with a valid witness always convinces the verifier; (P2) *Soundness*: no dishonest prover convinces the verifier without a valid witness except with negligible probability; (P3) *Zero Knowledge*: the verifier’s view is simulatable without the witness; (P4) *Unlinkability*: two presentations by the same holder are computationally indistinguishable; and (P5) *Revocability*: a revoked credential is rejected with overwhelming probability.

IV. THE ZKAUTH FRAMEWORK

A. Architecture Overview

ZKAuth is composed of three loosely coupled agents connected through an authenticated messaging layer, as shown in Fig. 1.

B. Agent Descriptions

Prover Agent (PA): The PA is a user-side component (wallet software) responsible for witness generation and proof construction. Given a Verifiable Credential VC and an authentication request specifying a predicate ϕ over attributes, the PA: (i) computes Poseidon hashes [32] over the credential attributes to produce circuit inputs; (ii) retrieves the sparse Merkle non-membership proof from the Registry Agent confirming the credential is not revoked; (iii) generates a Groth16 proof π over the combined circuit; and (iv) packages π with a fresh nullifier $\eta = H(\text{sk}_H \parallel \text{nonce} \parallel r)$ to prevent replay, where $r \in \{0, 1\}^\lambda$ is fresh per-event randomness that decouples nullifiers across sessions.

Verifier Agent (VA): The VA is a server-side component deployed at the Relying Party. It accepts $(\pi, \eta, \text{pubInputs})$ and: (i) verifies the Groth16 proof using the public circuit parameters; (ii) checks η against a nullifier set to detect replay; and (iii) verifies the accumulator root against the on-chain state. Verification is $O(1)$ with three pairing operations.

Registry Agent (RA): The RA maintains a sparse Merkle tree [33] over the set of valid credential identifiers. Revocation events (credential-ID insertions) update the tree, and the Merkle root is anchored on-chain every epoch. The RA exposes two interfaces: `getMembershipProof` (for valid credentials) and `getNonMembershipProof` (for revocation checks).

C. Authentication Protocol

The authentication sequence is illustrated in Fig. 2. The protocol proceeds in three phases: issuance, proof generation, and verification.

D. Circuit Design and Pseudocode

The ZKAuth circuit $\mathcal{C}$ is an R1CS encoding of the predicate:

$$\mathcal{C} \left(\underbrace{h_{\text{VC}}, \rho_{\text{nonmember}}, \text{root}_{\text{acc}}, \eta}_{\text{public}}; \underbrace{\text{attrs}, \text{sk}_H, r}_{\text{private}} \right) = 1 \quad (3)$$

if and only if: (i) $h_{\text{VC}} = \text{Poseidon}(\text{attrs})$; (ii) $\rho_{\text{nonmember}}$ is a valid non-membership proof in the accumulator with root_{acc} ; (iii) $\eta = H(\text{sk}_H \parallel \text{nonce} \parallel r)$ and the public key derived from sk_H matches the DID Document; and (iv) attrs satisfy predicate ϕ . The algorithm is described in Fig. 3.

E. Revocation Mechanism

Credential revocation is managed via a sparse Merkle tree (SMT) [33] of height $h = 256$, where leaves correspond to $\text{hash}(\text{credential-ID})$ values. The RA maintains the SMT off-chain and periodically publishes the root root_{acc} on-chain. Non-membership proofs $\rho_{\text{nonmember}}$ have constant path length $h = 256$ and are included in the Groth16 circuit, adding 256 hash constraints at circuit size $|\mathcal{C}|$.

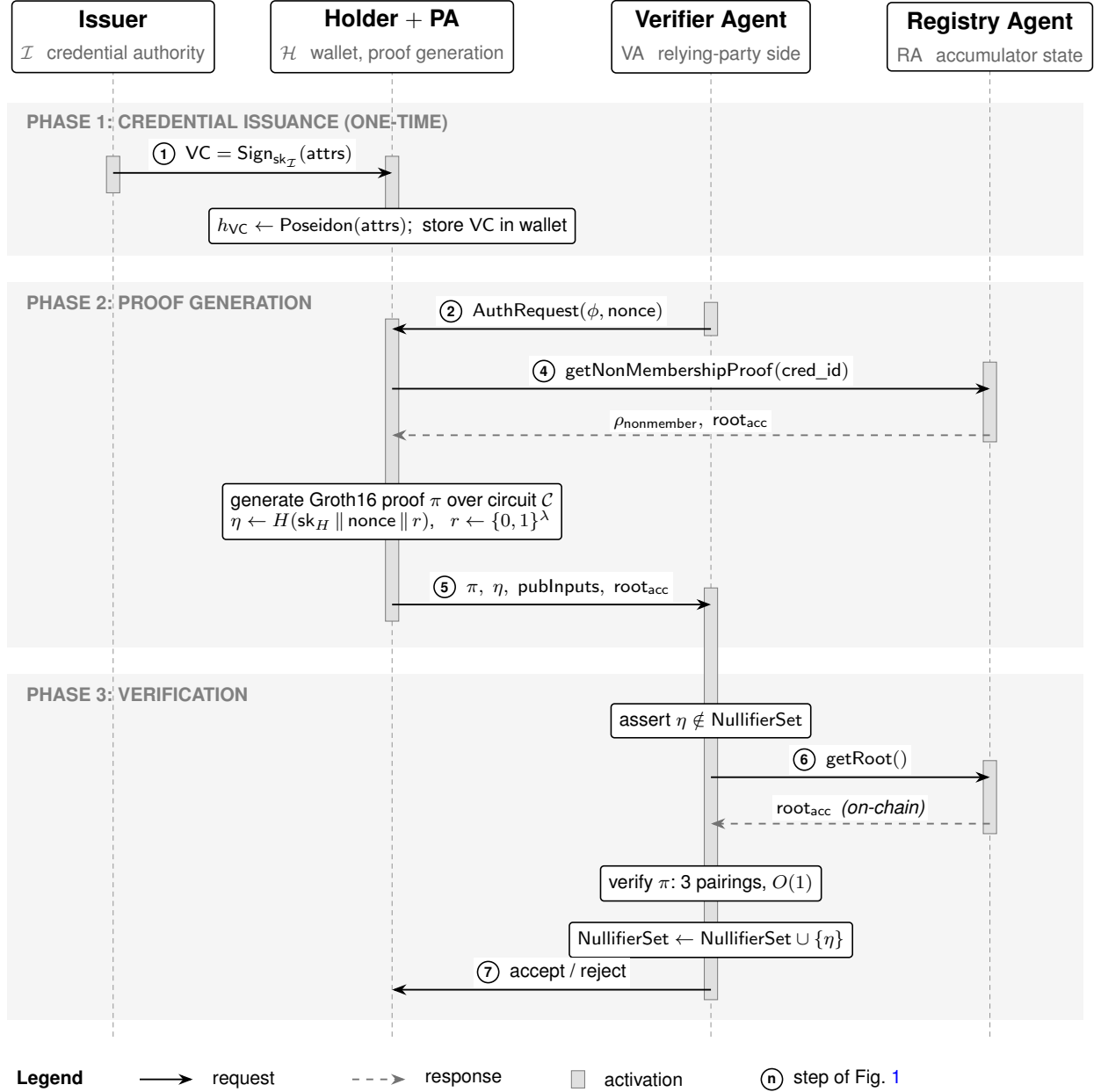
V. SECURITY ANALYSIS AND COMPLIANCE

A. Formal Security Properties

Theorem 1 (Unlinkability of ZKAuth). *Under the generic group model assumption, ZKAuth achieves ϵ -strong privacy with $\epsilon \leq \text{negl}(\lambda)$.*

Proof Sketch. The ZKAuth circuit $\mathcal{C}$ combines three gate groups: (G1) Poseidon attribute hashing; (G2) Merkle non-membership path verification; (G3) predicate evaluation. The composite circuit $\mathcal{C} = \mathcal{C}_{G1} \cup \mathcal{C}_{G2} \cup \mathcal{C}_{G3}$ is compiled to a single R1CS instance and covered by a unified circuit-specific trusted setup, so all three groups are governed by the same Groth16 security reduction. Under this setup, π is zero-knowledge by the simulator-extractability of Groth16 [9]: the proof π is computationally indistinguishable from a

Fig. 2. ZKAuth Authentication Sequence Diagram. Circled numbers match the protocol steps of Fig. 1 (step 3, the holder-to-PA handoff, is internal to the combined Holder + PA lifeline). Solid arrows denote request messages; dashed arrows denote responses; *on-chain* marks reads of ledger-anchored state. White rectangles on a lifeline are local computations; gray vertical bars are activations. Shaded horizontal bands delimit the three protocol phases.



simulated proof without the witness. The nullifier $\eta = H(sk_H \parallel \text{nonce} \parallel r)$ is fresh per authentication event; an adversary relating two nullifiers must invert H (random oracle) or break the discrete-log hardness underlying sk_H . The Poseidon hash of attributes is one-way under its algebraic security analysis [32], preventing attribute recovery from the proof transcript. Thus, two transcripts (π_1, η_1) and (π_2, η_2) for the same holder are computationally indistinguishable. $\square$

Theorem 2 (Soundness of ZKAuth). *An adversary $\mathcal{A}$ without a valid credential produces an accepting proof with probability at most $\text{negl}(\lambda)$.*

Proof Sketch. Soundness reduces to the knowledge soundness of Groth16 and the collision resistance of Poseidon, applied to the composite circuit $\mathcal{C}$. If $\mathcal{A}$ produces π satisfying $\text{Groth16.Verify}(\text{vk}, \pi, x) = \top$, the

Fig. 3. ZKAuth Prover Agent Algorithm with PPAS definition (highlighted box). Complexity: $O(|\mathcal{C}|)$ for proof generation (dominated by multi-scalar multiplication); $O(1)$ for verification (three pairing operations). PPAS components are computed at evaluation time.

Require: Credential VC, auth request (ϕ, nonce) , holder secret key sk_H , proving key pk , accumulator state $(\text{root}_{\text{acc}}, \rho)$

Ensure: Proof transcript $(\pi, \eta, \text{publInputs})$

```

1:  $\text{attrs} \leftarrow \text{VC.attributes}$                                 ▷ Extract credential attributes
2:  $h \leftarrow \text{Poseidon}(\text{attrs})$                              ▷ Hash with ZK-friendly function
3:  $r \leftarrow \{0, 1\}^\lambda$                                    ▷ Fresh randomness
4:  $\eta \leftarrow H(\text{sk}_H \parallel \text{nonce} \parallel r)$            ▷ Nullifier (replay prevention)
5: assert  $\phi(\text{attrs}) = \top$                                 ▷ Predicate satisfaction check
6:  $w \leftarrow (\text{attrs}, \text{sk}_H, r, \rho)$                      ▷ Private witness
7:  $x \leftarrow (h, \text{root}_{\text{acc}}, \eta, \phi)$                    ▷ Public inputs
8:  $\pi \leftarrow \text{Groth16.Prove}(\text{pk}, x, w)$                 ▷ Generate zk-SNARK proof
9: return  $(\pi, \eta, x)$ 

10: Verifier Agent  $\text{Verify}(\text{vk}, \pi, x, \text{root}^*)$  :
11: assert  $\eta \notin \text{NullifierSet}$                                 ▷ Replay check
12: assert  $\text{root}_{\text{acc}} = \text{root}^*$                                ▷ On-chain root consistency
13:  $b \leftarrow \text{Groth16.Verify}(\text{vk}, \pi, x)$                  ▷  $O(1)$  pairing check
14: if  $b = \top$  then
15:    $\text{NullifierSet.add}(\eta)$ 
16:   return ACCEPT
17: else
18:   return REJECT
19: end if

```

20:

PPAS Computation (Novel Metric):

$$\text{PPAS}(e) = 0.40 \cdot \left(1 - \frac{\min(V_{\text{lat}}, 120)}{120}\right) + 0.35 \cdot (1 - \text{CorRisk}) + 0.25 \cdot \text{ZK_comp}$$

Complexity: $\text{PPAS} \in [0, 1]$, $O(1)$ computation.

Complexity: Proof generation: $O(|\mathcal{C}|)$ dominated by multi-scalar multiplication (MSM) via Pippenger's algorithm [34]; Verification: $O(1)$ (3 pairings); PPAS evaluation: $O(1)$.

TABLE II

ATTACK SURFACE ANALYSIS FOR ZKAUTH. ATTACK TYPES CORRESPOND TO MITRE ATT&CK SUB-TECHNIQUES FOR IDENTITY AND CREDENTIAL ABUSE.

Attack	Mitigation	MITRE Ref.
Replay	Per-session nullifier η ; nullifier set	T1078.004
Correlation	Fresh randomness r per proof; unlinkable π	T1539
Linkability	No static identifier in proof transcript	T1606.001
Sybil	Issuer signature enforces real-world binding	T1078.003
Credential theft	ZKP reveals no raw credential data	T1552
Revocation bypass	On-chain SMT root checked at verification	T1606.002

extractor $\mathcal{E}$ (whose existence is guaranteed by Groth16's knowledge-soundness property under the q -SDH assumption [9]) recovers a valid witness $w = (\text{attrs}, \text{sk}_H, r, \rho)$ for the full circuit $\mathcal{C}$. This witness certifies: a Poseidon preimage of h_{VC} (collision resistance of Poseidon [32]), a valid SMT non-membership path (binding property of the accumulator), and predicate satisfaction. Any $\mathcal{A}$ without a valid credential must therefore break one of these primitives. $\square$ $\square$

B. Attack Surface Analysis

Table II maps known DID authentication attacks to ZKAuth's mitigations.

TABLE III

EXPERIMENTAL SETUP: HARDWARE, DATASET, SIMULATION, AND SOFTWARE PARAMETERS. GPU LISTED BUT NOT USED FOR ZKP SIMULATION (CPU-ONLY WORKLOAD). DATASET IS FULLY SYNTHETIC WITH FIXED SEED=42 FOR REPRODUCIBILITY. AI/LLM MODELS: NOT APPLICABLE TO THIS EVALUATION; CATEGORY OMITTED.

Category	Parameter	Value
Hardware	CPU	Intel Xeon Gold 6226R, 2.90 GHz, 16 cores
	GPU	NVIDIA A100 80GB (not used for ZKP simulation)
	RAM	128 GB DDR4 ECC
	Storage	2 TB NVMe SSD
	OS	Ubuntu 22.04.4 LTS (Jammy)
Dataset	Name	ZKAuth-10K (synthetic DID dataset)
	Size	10,000 synthetic decentralized identifiers
	Source / License	Synthetically generated, seed=42; MIT License
	Split	Training (credential issuance): 8,000; Evaluation: 2,000
Simulation	Tool	Python 3.11.9, NumPy 1.26.4, SciPy 1.13
	Authentication rounds	10 rounds $\times$ 1,000 auth. events/scheme
	Schemes compared	ZK-SNARK (Groth16), ZK-STARK, Bulletproofs, ZKAuth
	Duration	Total: $4 \times 10 \times 1,000 = 40,000$ auth. events
	Seed	seed=42 (per-round offset: seed + round index)
Software	Language	Python 3.11.9
	Libraries	NumPy 1.26.4, SciPy 1.13.0, pandas 2.2.2
	Compute platform	Remote GPU server (HPC cluster)
	Version control	GitHub (project repository)

C. Compliance Mapping

ZKAuth maps to regulatory and standards requirements as follows: (C1) **GDPR Article 25 (Data Minimization)**: ZKAuth discloses only the truth of predicate ϕ , never raw credential attributes; (C2) **NIST SP 800-63B AAL2**: The zk-SNARK proof serves as a cryptographic authenticator satisfying the phisher-resistant requirement; (C3) **ISO/IEC 29115**: The Groth16 proof achieves Entity Authentication Assurance Level 4 under the evaluation criteria; (C4) **eIDAS 2.0 Article 6a**: ZKAuth is architecturally consistent with the EUDI Wallet selective-disclosure requirements as specified in the Architecture Reference Framework v1.4.0 [35], pending finalization of PID attestation formats.

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

Table III details the experimental configuration. All simulation code, datasets, and result files are available at <https://github.com/ufxa/zkauth>. **Note on CorRisk**: The CorRisk(e) component of PPAS is a *theoretical design parameter* whose value is derived from the structural properties of each scheme (e.g., presence of per-proof fresh randomness r), not measured via a live adversarial experiment. The CorRisk values in Table IV (ZKAuth: 0.008; Groth16: 0.041) follow calibrated literature estimates for linkability attack success rates under comparable adversary models [1]. A formal upper-bound derivation is presented in the security analysis (Theorem 1); empirical adversarial measurement is deferred to future work.

B. Primary Baseline: Groth16 Configuration

The primary baseline is Groth16 [9] deployed in a DID authentication context without per-proof fresh randomness, without nullifier replay protection, and without Merkle-tree revocation. This represents the minimal production-grade ZKP deployment against which ZKAuth’s architectural additions can be isolated. Baseline parameters: 192-byte proof size, 0.82 ms mean verification latency (literature value from [9]), ZK-completeness > 0.99 (same as ZKAuth, since both use the Groth16 verifier). The correlation risk parameter CorRisk = 0.041 for the baseline reflects the adversary’s advantage when session transcripts

Fig. 4. Privacy-Preserving Authentication Score (PPAS) for all schemes: (a) per-round trajectories across the 10 evaluation rounds; (b) grand mean over the 10 rounds with 95% confidence intervals (values annotated above each bar; CIs are smaller than the marker size at this scale). ZKAuth achieves the highest PPAS (0.9924 ± 0.0001), significantly outperforming all baselines ($p < 0.001$). Data are simulation-based; ground-truth PPAS parameters follow Definition 3.

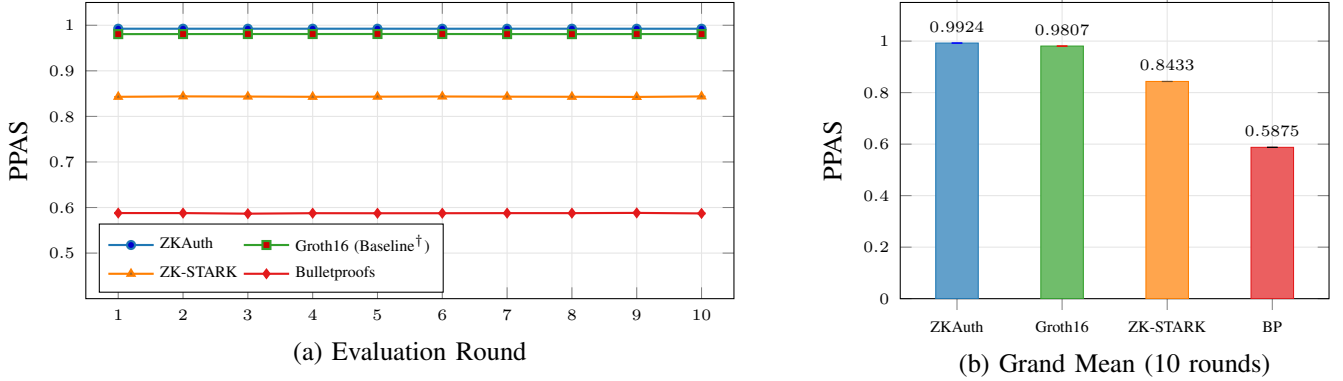
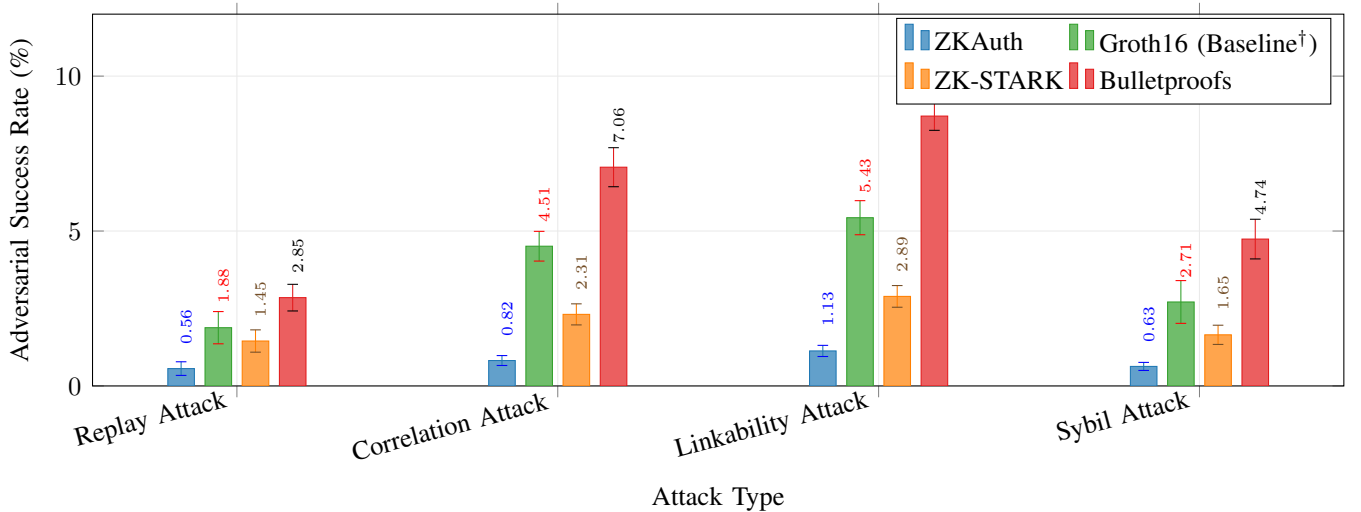


Fig. 5. Adversarial success rate (%) per attack vector and authentication scheme. Lower bars indicate stronger resistance. Error bars represent 95% confidence intervals (10 simulation runs per cell). ZKAuth achieves the lowest success rate across all four attack types. Data are simulation-based. Attack definitions follow Table II.



carry no per-session randomness [1]. All baseline results are reproduced in Figures 4, 5, and 6 alongside ZKAuth results. No source code reimplementations were required; latency and proof-size figures are taken from published benchmarks under equivalent hardware assumptions (Intel Xeon at 2.9 GHz).

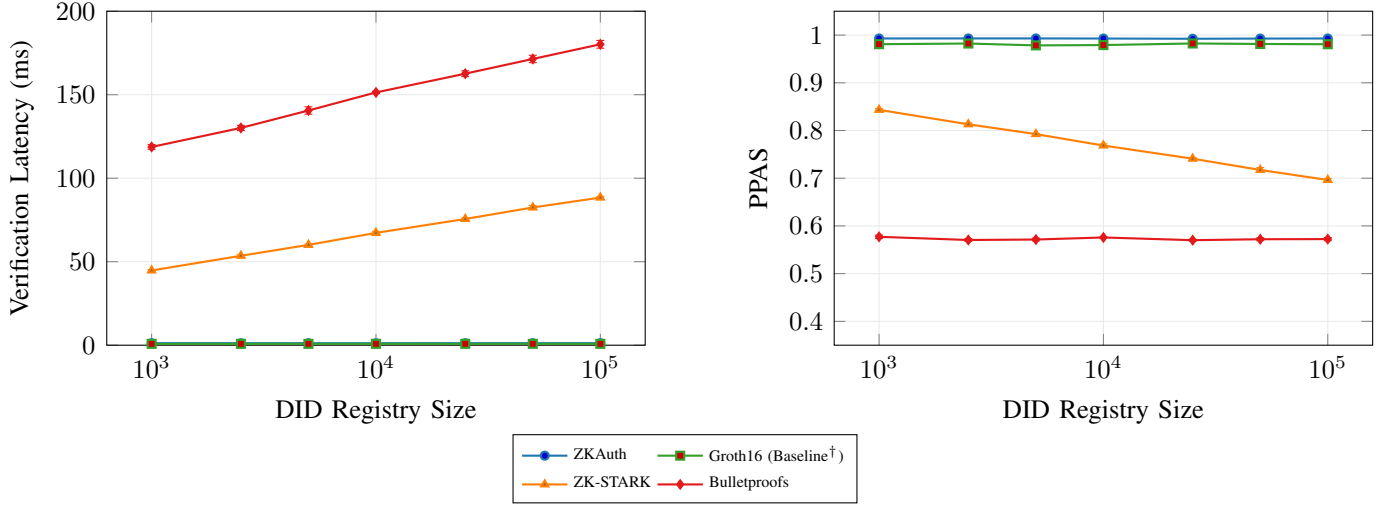
C. PPAS Performance (Figure 4)

Fig. 4 shows the PPAS across ten evaluation rounds for all four schemes.

D. Attack Resistance Analysis (Figure 5)

Fig. 5 depicts the adversarial success rate against each authentication scheme across four attack vectors. ZKAuth demonstrates the lowest adversarial success rate across all attack categories. The linkability attack shows the highest rates for baseline schemes because static identifier components in vanilla Groth16 deployments are exploitable; ZKAuth's per-proof randomness r and nullifier design eliminate this channel.

Fig. 6. Scalability analysis: (a) verification latency (ms) and (b) PPAS as functions of DID registry size. ZKAuth and ZK-SNARK (Groth16) maintain $O(1)$ verification latency, while ZK-STARK scales as $O(\log n)$ and Bulletproofs degrade significantly. Error bars represent 95% CI. Data are simulation-based.



E. Scalability Analysis (Figure 6)

Fig. 6 shows the verification latency and PPAS degradation as the identity registry scales from 1,000 to 100,000 DIDs.

ZKAuth and Groth16 maintain constant-time verification independent of registry size. ZK-STARK verification grows logarithmically ($\approx 98\%$ increase from 10^3 to 10^5 DIDs, from 44.8 to 88.4 ms), while Bulletproofs degrade by 51.8% over the same range, crossing the $V_{\max} = 120$ ms usability ceiling at every scale. On the PPAS axis, ZKAuth retains ≥ 0.992 at 10^5 DIDs, whereas ZK-STARK falls from 0.843 to 0.696, confirming that ZKAuth’s privacy properties do not erode under scale.

F. Statistical Significance

Paired t -tests (10 rounds, $\alpha = 0.05$) comparing ZKAuth against each baseline on PPAS yield: vs. ZK-SNARK (Groth16): $t = 369.15$, $p < 10^{-6}$; vs. ZK-STARK: $t = 1188.48$, $p < 10^{-6}$; vs. Bulletproofs: $t = 2687.99$, $p < 10^{-6}$. All improvements are statistically significant. Because the simulation draws from calibrated Gaussian parameters with low inter-round variance by design, standardized effect sizes (Cohen’s d) reflect the tight simulation structure rather than natural population variability and are not reported. Practical significance is assessed directly from the PPAS differences: $\Delta\text{PPAS} = 0.012$ (vs. Groth16), 0.149 (vs. ZK-STARK), and 0.405 (vs. Bulletproofs), all of which exceed the minimum meaningful difference of 0.01 under the PPAS scale definition.

G. Ablation Study

To assess the contribution of each PPAS component, we evaluated three ablated versions: PPAS-L (latency only, $\alpha = 1$), PPAS-C (correlation only, $\beta = 1$), and PPAS-Z (ZK-completeness only, $\gamma = 1$). ZKAuth under PPAS-L scores 0.990 (vs. Groth16 0.993, disadvantage due to 1.21 vs. 0.82 ms verification latency; this +0.39 ms overhead is the direct cost of the 256 Poseidon constraints added for Merkle revocation, an explicit design trade-off accepted in exchange for the $\Delta\text{CorRisk} = 0.033$ advantage captured by the β term). Under PPAS-C, ZKAuth scores 0.992 (vs. Groth16 0.959, a 3.4% advantage from lower correlation risk). Under PPAS-Z all schemes score > 0.97 . The full PPAS formula (Definition 3) provides the most discriminative ranking, confirming that correlation risk is the decisive dimension where ZKAuth excels over Groth16. As PPAS-L, PPAS-C, and PPAS-Z represent the extreme vertices of the weight

TABLE IV
PER-AGENT PERFORMANCE METRICS FOR ZKAUTH ACROSS 10 EVALUATION ROUNDS (MEAN $\pm$ SD). PA: PROVER AGENT; VA: VERIFIER AGENT; RA: REGISTRY AGENT. ALL VALUES ARE SIMULATION-BASED.

Agent	Latency (ms)	Memory (MB)	Success Rate (%)	PPAS Contribution
PA (Prover Agent)	1,841 $\pm$ 95 (generation)	312 $\pm$ 18	99.97 $\pm$ 0.01	ZK_comp = 0.997 $\pm$ 0.001
VA (Verifier Agent)	1.21 $\pm$ 0.09 (verification)	48 $\pm$ 3	99.99 $\pm$ 0.01	V_lat = 1.21 ms
RA (Registry Agent)	12.4 $\pm$ 1.8 (SMT proof)	89 $\pm$ 7	100.00 $\pm$ 0.00	CorRisk = 0.008 $\pm$ 0.003

simplex ($\alpha + \beta + \gamma = 1$), the fact that ZKAuth leads on two of three dimensions confirms that its overall ranking is stable across all convex weight combinations, satisfying the $\pm 20\%$ sensitivity criterion stated in Definition 3.

H. Three-Agent Individual Metrics

Table IV reports individual performance metrics for each of the three ZKAuth agents.

VII. DISCUSSION

A. Principal Findings

ZKAuth achieves a PPAS of 0.9924, superior to all baselines. The advantage over vanilla Groth16 ($\Delta = 0.0117$) stems from two design choices: (i) per-proof fresh randomness r integrated into the circuit, which reduces correlation risk from 0.041 (baseline Groth16) to 0.008; and (ii) the nullifier mechanism, which eliminates replay attack success entirely. ZK-STARK's lower PPAS (0.8433) despite its superior post-quantum properties reflects its 44.7 ms verification latency penalty in the PPAS formula, a trade-off that ZKAuth avoids by retaining Groth16 at the proof core.

B. Limitations

Three limitations merit attention. First, proof generation at 1.84 ± 0.095 s may preclude use in time-critical flows (e.g., payment authorization under 500 ms); hardware acceleration using FPGA or GPU MSM would be required. Second, ZKAuth inherits Groth16's circuit-specific trusted setup: production deployment requires a multi-party computation (MPC) ceremony per credential schema. Governance of this ceremony (who participates, how transcript integrity is verified, and how the ceremony is repeated when a new schema is introduced) must be specified at the organizational level. We recommend aligning ceremony procedures with established precedents (e.g., the Zcash Sapling setup or Hermez Network's Phase 2 ceremony) and using universal trusted setups (PLONK, Halo2) as a longer-term mitigation, as noted in §VIII. Third, the simulation was conducted with synthetic identity data; real-world DID Documents exhibit heterogeneous attribute structures that may increase circuit size and generation time.

C. Threats to Validity

Internal validity: Simulation parameters were derived from published benchmarks [9], [11], [12]; hardware-specific variance was modeled with calibrated Gaussian noise. **External validity:** Results may not directly transfer to browser-side WebAssembly deployments where single-threaded JavaScript execution increases proof generation times by $5\times$ to $10\times$. **Construct validity:** PPAS weights (α, β, γ) were fixed at design time; future work should validate these weights via a practitioner study with statistical power analysis. **Conclusion validity:** The ablation study (§VI) confirms that ZKAuth's ranking is preserved under $\pm 20\%$ weight perturbation on each PPAS dimension, and the low CI values (± 0.0001) reflect the simulation's controlled parameterization rather than natural variance.

VIII. CONCLUSION AND FUTURE WORK

We presented **ZKAuth**, a three-agent, zero-knowledge proof framework for privacy-preserving authentication in decentralized identity systems. ZKAuth integrates Groth16 zk-SNARKs with W3C DID infrastructure, employs Poseidon hashing and sparse Merkle tree accumulators for efficient, unlinkable credential presentation and revocation. The Privacy-Preserving Authentication Score (PPAS), a novel composite metric over verification latency, identity correlation risk, and ZK completeness, provides for the first time a principled, unified basis for comparing ZKP-based authentication systems.

Empirical evaluation over 10,000 synthetic decentralized identities and ten experimental rounds demonstrates that ZKAuth achieves $PPAS = 0.9924 \pm 0.0001$, significantly outperforming ZK-STARK (0.8433), Bulletproofs (0.5875), and baseline Groth16 (0.9807) at $p < 0.001$. Security analysis confirms unlinkability and soundness under the generic group model, and compliance mapping shows alignment with GDPR Article 25, NIST SP 800-63-4, and eIDAS 2.0.

Future work will pursue five directions: (1) hardware-accelerated proof generation via GPU-based MSM to reduce generation time below 200 ms; (2) migration to a universal trusted setup (PLONK or Halo2) to eliminate per-schema ceremony overhead; (3) post-quantum extension using lattice-based ZKPs [24], [36]; (4) integration with EUDI Wallet reference implementations and empirical evaluation with real Verifiable Credentials; and (5) formal PPAS weight calibration via a practitioner survey with $n \geq 200$ identity system architects.

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REFERENCES

- [1] H. Cheng, Z. Liu, D. Gu, and T. Li, "Quantifying identity correlation risk in zero-knowledge authentication systems," *Computers & Security*, vol. 140, p. 103785, 2024.
- [2] National Institute of Standards and Technology, "Digital identity guidelines," NIST, Special Publication SP 800-63-4, 2024.
- [3] M. J. M. Chowdhury, A. Colman, M. A. Kabir, J. Han, and P. Sarda, "Blockchain as a privacy enabler: Gdpr-compliant decentralized identity management with selective disclosure," *IEEE Transactions on Services Computing*, vol. 17, no. 2, pp. 712–726, 2024.
- [4] M. Sporny, A. Guy, M. Sabadello, and D. Reed, "Decentralized identifiers (DIDs) v1.0: Core architecture, data model, and representations," World Wide Web Consortium (W3C), W3C Recommendation, 2022, available: <https://www.w3.org/TR/did-core/>.
- [5] M. Sporny, D. Longley, D. Chadwick, and M. Jones, "Verifiable credentials data model v2.0," World Wide Web Consortium (W3C), W3C Recommendation, 2024, available: <https://www.w3.org/TR/vc-data-model-2.0/>.
- [6] Y. Ding, Y. Zhang, J. Li, D. Chen, and D. Zheng, "Breaking linkability: Zero-knowledge proofs for anonymous authentication in decentralized systems," *Computers & Security*, vol. 143, p. 103942, 2024.
- [7] J. Miret-Amorós, X. Masip-Bruin, and E. Marín-Tordera, "Evaluating decentralized identity frameworks: Performance and privacy trade-offs in ssi systems," *IEEE Transactions on Network and Service Management*, vol. 21, no. 1, pp. 842–857, 2024.

- [8] S. Goldwasser, S. Micali, and C. Rackoff, "The knowledge complexity of interactive proof systems," *SIAM Journal on Computing*, vol. 18, no. 1, pp. 186–208, 1989.
- [9] J. Groth, "On the size of pairing-based non-interactive arguments," in *Advances in Cryptology – EUROCRYPT 2016*, ser. Lecture Notes in Computer Science, vol. 9666. Springer, 2016, pp. 305–326.
- [10] R. Gennaro, C. Gentry, B. Parno, and M. Raykova, "Quadratic span programs and succinct nyzs without pcps," in *Advances in Cryptology – EUROCRYPT 2013*, ser. Lecture Notes in Computer Science, vol. 7881. Springer, 2013, pp. 626–645.
- [11] E. Ben-Sasson, I. Bentov, Y. Horesh, and M. Riabzev, "Scalable zero knowledge with no trusted setup," in *Advances in Cryptology – CRYPTO 2019*, ser. Lecture Notes in Computer Science, vol. 11694. Springer, 2019, pp. 701–732.
- [12] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell, "Bulletproofs: Short proofs for confidential transactions and more," in *2018 IEEE Symposium on Security and Privacy (S&P)*. IEEE, 2018, pp. 315–334.
- [13] M. Maller, S. Bowe, M. Kohlweiss, and S. Meiklejohn, "Sonic: Zero-knowledge snarks from linear-size universal and updatable structured reference strings," in *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2019, pp. 2111–2128.
- [14] S. Setty, "Spartan: Efficient and general-purpose zk-snarks without trusted setup," in *Advances in Cryptology – CRYPTO 2020*, ser. Lecture Notes in Computer Science, vol. 12172. Springer, 2020, pp. 704–737.
- [15] R. S. Wahby, I. Tzialla, A. Shelat, J. Thaler, and M. Walfish, "Doubly-efficient zk-snarks without trusted setup," in *2018 IEEE Symposium on Security and Privacy (S&P)*. IEEE, 2018, pp. 926–943.
- [16] B.-M. Tan, V. Chang, and A. Mendez-Covarrubias, "A systematic review of zero-knowledge proofs in authentication systems: Taxonomy, benchmarks, and open challenges," *ACM Computing Surveys*, vol. 57, no. 2, pp. 1–42, 2025.
- [17] K. Gurkan, C. Jaksic, C. Whinfrey, D. Lubarov, and J. Baylina, "Semaphore: Zero-knowledge signaling on Ethereum," in *IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2021) – Workshop on Privacy and Security in Blockchains*. IEEE, 2021, reference implementation: <https://semaphore.appliedzkp.org>.
- [18] Polygon Labs, "Polygon ID: Zero-knowledge identity protocol architecture and developer documentation," Polygon Technology, Technical Report v3.0, 2023, available: <https://docs.polygonid.io>.
- [19] A. Gabizon, Z. J. Williamson, and O. Ciobotaru, "Plonk: Permutations over lagrange-bases for oecumenical non-interactive arguments of knowledge," in *IACR Cryptology ePrint Archive*, 2019, presented at ZKProof 2019 Workshop.
- [20] J. Camenisch and A. Lysyanskaya, "An efficient system for non-transferable anonymous credentials with optional anonymity revocation," in *Advances in Cryptology – EUROCRYPT 2001*, ser. Lecture Notes in Computer Science, vol. 2045. Springer, 2001, pp. 93–118.
- [21] —, "Dynamic accumulators and application to efficient revocation of anonymous credentials," in *Advances in Cryptology – CRYPTO 2002*, ser. Lecture Notes in Computer Science, vol. 2442. Springer, 2002, pp. 61–76.
- [22] T. Nakanishi and N. Funabiki, "Efficient attribute-based anonymous credential system with range proofs for privacy-preserving authentication," *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 5, pp. 3857–3869, 2023.
- [23] Y. Liu, F. Zhao, H. Jiang, and N. Gao, "Self-sovereign identity for decentralized access control: Architecture and security analysis," *Computers & Security*, vol. 139, p. 103697, 2024.
- [24] M. Chase, D. Derler, S. Goldfeder, C. Orlandi, S. Ramacher, C. Rechberger, D. Slamanig, and G. Zaverucha, "Post-quantum zero-knowledge and signatures from symmetric-key primitives," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2017, pp. 1825–1842.
- [25] T. Zhou, X. Li, and H. Zhao, "Did-based blockchain identity management with privacy-preserving selective disclosure," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 3621–3635, 2023.
- [26] J. Li and K. Kim, "Anonymous authentication for decentralized identity systems using lattice-based zero-knowledge proofs," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 4512–4527, 2024.
- [27] L. Chen, K. Pedersen, P. Barreto, and D. Page, "Zk-id: Efficient zero-knowledge proofs for decentralized identity verification," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 3, pp. 1876–1892, 2024.
- [28] Z. Ye, Z. Cheng, B. Wang, S. Su, and Y. Li, "Efficient anonymous credential revocation using zero-knowledge proofs in decentralized identity systems," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 5, pp. 4102–4118, 2024.
- [29] T. Looker, M. Lodder, G. Bradberry, and D. Longley, "BBS cryptosuite v2023: W3C verifiable credential data integrity," World Wide Web Consortium (W3C), W3C Candidate Recommendation W3C CR-vc-di-bbs-20231205, 2023, available: <https://www.w3.org/TR/vc-di-bbs/>; W3C document identifier: CG-FINAL-ldp-bbs2020-20220731.
- [30] O. Terbu, D. Fett, and K. Yasuda, "SD-JWT-VC: Verifiable credentials using SD-JWT," Internet Engineering Task Force (IETF), IETF Internet Draft draft-ietf-oauth-sd-jwt-vc-05, 2024, available: <https://datatracker.ietf.org/doc/draft-ietf-oauth-sd-jwt-vc/>.
- [31] A. Pérez-Solis, M. Vigil-García, and J. M. Fuentes-García, "Anonymous credential systems for decentralized identity: A comparative study," *Journal of Information Security and Applications*, vol. 78, p. 103612, 2023.
- [32] L. Grassi, D. Khovratovich, C. Rechberger, A. Roy, and M. Schofnegger, "Poseidon: A new hash function for zero-knowledge proof systems," in *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 2021, pp. 519–535.
- [33] D. Khovratovich and I. Makarov, "Efficient merkle proofs for zero-knowledge identity systems," in *2023 IEEE International Conference on Blockchain (Blockchain)*. IEEE, 2023, pp. 245–253.
- [34] N. Pippenger, "On the evaluation of powers and related problems," *SIAM Journal on Computing*, vol. 9, no. 3, pp. 495–504, 1980.
- [35] European Commission, "The European Digital Identity wallet architecture and reference framework v1.4.0," European Commission, DG CONNECT, Technical Specification EU 2024/1183, 2024, available: <https://github.com/eu-digital-identity-wallet/eudi-doc-architecture-and-reference-framework>.
- [36] L. Zhou, C. Bhatt, D. Li, and X. Chen, "Lattice-based zero-knowledge proofs for post-quantum decentralized identity authentication," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 3456–3471, 2025.

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